

Ex 13.8

maxima & minima of function by two variables with continuous ~~2nd~~ 2nd order

Let f be a function of two variables with continuous 2nd order partial derivatives in some disk centered at a critical point (x_0, y_0) & let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

- ① if $D > 0$ & $f_{xx}(x_0, y_0) > 0$, then f has a relative minima at (x_0, y_0)
- ② if $D > 0$ & $f_{xx}(x_0, y_0) < 0$, " " " " maxima at " "
- ③ if $D < 0$, then f has saddle point at (x_0, y_0)
- ④ if $D = 0$, then no conclusion can be drawn.

Example 3

Locate all relative extrema & saddle point of:

$$f(x, y) = 3x^2 - 2xy + y^2 - 8y$$

Sol: -

$$f_x = 6x - 2y, \quad f_{xx} = 6$$

$$f_y = -2x + 2y - 8, \quad f_{yy} = 2$$

$$\begin{aligned} 6x - 2y &= 0 \\ + -2x + 2y - 8 &= 0 \\ 4x &= 8 \\ x &= 2 \end{aligned}$$

$$\begin{aligned} 12 - 2y &= 0 \\ 12 &= 2y \\ y &= 6 \end{aligned}$$

$$\begin{aligned} D &= f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) \\ &= 6 \cdot 2 - (-2)^2 \\ &= 12 - 4 \\ &= 8 > 0 \end{aligned}$$

minima

Ex 13.8 Example 3, 4

Ques 3, 4 and 13

Q3,4 Complete the eq. and locate all absolute maxima and minima, if any, by inspection. Then check your answer by using calculus

Q3 $f(x,y) = 13 - 6x + x^2 + 4y + y^2$

$$f_x = -6 + 2x \Rightarrow 2x - 6, \quad f_{xx} = 2$$
$$f_y = 4y + 4, \quad f_{yy} = 2$$

$$\begin{array}{l} 2x - 6 = 0 \\ 4y + 4 = 0 \end{array} \quad f_{xy} = 0$$

$$\begin{array}{ll} 2x - 6 = 0 & 4y + 4 = 0 \\ 2x = 6 & 4y = -4 \\ x = \frac{6}{2} & y = -1 \end{array}$$

$$x = 3$$

$$y = -1$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$
$$D = 2 \cdot 2 - (0^2)$$
$$D = 4$$

$$4 > 0$$

minima.

$$\text{Qu)} \quad f(x, y) = 1 - 2x - x^2 + 4y - 2y^2$$

$$f'_x = -2 - 2x$$

$$f'_y = 4 - 4y$$

$$f_{xx} = -2$$

$$f_{yy} = -4$$

$$f_{xy} = 0$$

$$-2 - 2x = 0$$

$$-2x = 2$$

$$x = -1$$

$$4 - 4y = 0$$

$$-4y = -4$$

$$y = 1$$

$$D = f''_{xx}(x_0, y_0) f''_{yy}(x_0, y_0) - (f''_{xy}(x_0, y_0))^2$$

$$D = (-2)(-4) - 0$$

$$D = 8$$

$$8 > 0$$

relative maxima.

Q13)

$$f(x, y) = x^2 + y^2 + \frac{2}{xy}$$

$$f_x = 2x - \frac{2}{yx^2}, \quad f_{xx} = 2 + \frac{4}{x^3y}$$

$$f_y = 2y - \frac{2}{xy^2}, \quad f_{yy} = 2 + \frac{4}{xy^3}$$

$$f_{xy} = \frac{2}{x^2y^2}$$

$$2x - \frac{2}{yx^2} = 0$$

$$2y = \frac{2}{xy^2}$$

$$2x = \frac{2}{yx^2}$$

$$2y = \frac{2}{y^{1/3} \cdot y^2}$$

$$2x^3y = 2$$

$$x^3y = 1$$

$$x^3 = \frac{1}{y}$$

$$x = \sqrt[3]{\frac{1}{y}}$$

$$2y^3 = 2y^{1/3}$$

$$1 = y^{1/3} \cdot 3^{-1/3}$$

$$y^{1/3} = 1$$

$$y = (1^3)^{3 \cdot 1/3}$$

$$y = 1$$

$$x = \frac{1}{(y)^3}$$

$$x = \frac{1}{(1)^3}$$

$$x = 1$$

$$D = \left(2 + \frac{4}{x^3 y}\right) \left(2 + \frac{4}{x y^3}\right) - \frac{4}{(x^4 y^4)}$$

$$= \left(\frac{2x^3 y + 4}{x^3 y}\right) \left(\frac{2x y^3 + 4}{x y^3}\right) - \frac{4}{x^4 y^4}$$

$$= 4$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$= \frac{2 + \frac{4}{1^3 \cdot 1}}{1^3 \cdot 1} \cdot \frac{2 + \frac{4}{1 \cdot 1^3}}{1 \cdot 1^3} - \frac{4}{1^4 \cdot 1^4}$$

$$= 6 \cdot 6 - 4$$

$$= 36 - 4$$

$$= 32 > 0$$

relative minima.

Example 4

Locate all relative extrema and saddle points of

$$f(x, y) = 4xy - x^4 - y^4$$

Sol:-

$$f(x, y) = 4xy - x^4 - y^4$$

$$f_x = 4y - 4x^3$$

$$f_{xy} = 4$$

$$f_y = 4x - 4y^3$$

$$4y - 4x^3 = 0$$

$$4(y - x^3) = 0$$

$$y - x^3 = 0$$

$$y = x^3$$

or

$$x = y^3$$

$$4x - 4y^3 = 0$$

$$f_{xx} = -12x^2$$

$$f_{yy} = -12y^2$$

$$4y - 4y = 0$$

$$0 = 0$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

$$= (-12x^2)(-12y^2) - (4)^2$$

$$=$$

$$x=0$$

$$x^8 - 1 = 0$$

$$(x^2)^{3-\frac{1}{3}} = 0$$

$$x^2 - 1 = 0$$

$$x = (x^3)^3$$

$$x^9 - x = 0$$

$$x(x^8 - 1) = 0$$

$$x = 0, x = 1, x = -1$$

$$y = 0, y = 1, y = -1$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

So,
Critical
points

	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$
$(0, 0)$	0	0	4
$(1, 1)$	-12	-12	4
$(-1, -1)$	-12	-12	4

Thus D becomes

$$D = -16$$

$$D = 128$$

$$D = 128$$